

## Day2\_HomeAssignment1

>\_MONGOSH

```
> db.createCollection("patients")
< { ok: 1 }
> db.patients.insertMany([
  { patientName: "Arun", doctor: "Dr. Ravi", department: "Cardiology", age: 45,
    treatmentCost: 20000, daysAdmitted: 5 },
  { patientName: "Meena", doctor: "Dr. Kumar", department: "Neurology", age: 50,
    treatmentCost: 30000, daysAdmitted: 7 },
  { patientName: "Raj", doctor: "Dr. Ravi", department: "Cardiology", age: 60,
    treatmentCost: 15000, daysAdmitted: 3 },
  { patientName: "Divya", doctor: "Dr. Mehta", department: "Orthopedics", age: 35,
    treatmentCost: 10000, daysAdmitted: 2 },
  { patientName: "Kiran", doctor: "Dr. Kumar", department: "Neurology", age: 40,
    treatmentCost: 25000, daysAdmitted: 6 },
  { patientName: "Sneha", doctor: "Dr. Mehta", department: "Orthopedics", age: 30,
    treatmentCost: 8000, daysAdmitted: 2 },
  { patientName: "Vijay", doctor: "Dr. Ravi", department: "Cardiology", age: 55,
    treatmentCost: 22000, daysAdmitted: 4 },
  { patientName: "Priya", doctor: "Dr. Kumar", department: "Neurology", age: 48,
    treatmentCost: 27000, daysAdmitted: 5 },
  { patientName: "Anil", doctor: "Dr. Mehta", department: "Orthopedics", age: 52,
    treatmentCost: 12000, daysAdmitted: 3 },
  { patientName: "Riya", doctor: "Dr. Ravi", department: "Cardiology", age: 29,
    treatmentCost: 18000, daysAdmitted: 4 }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a115fab09c9f7fe9b192d66'),
    '1': ObjectId('6a115fab09c9f7fe9b192d67'),
```

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```
> db.patients.aggregate([
  {$match: { treatmentCost: { $gt: 20000 } }},
  {$project: {
    _id: 0,
    patientName: 1,
    department: 1}}
])
< {
  patientName: 'Meena',
  department: 'Neurology'
}
{
  patientName: 'Kiran',
  department: 'Neurology'
}
{
  patientName: 'Vijay',
  department: 'Cardiology'
}
{
  patientName: 'Priya',
  department: 'Neurology'
}
```

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```
> db.patients.aggregate([
  {$sort: { treatmentCost: -1 }}
])
< {
  _id: ObjectId('6a115fab09c9f7fe9b192d67'),
  patientName: 'Meena',
  doctor: 'Dr. Kumar',
  department: 'Neurology',
  age: 50,
  treatmentCost: 30000,
  daysAdmitted: 7
}
{
  _id: ObjectId('6a115fab09c9f7fe9b192d6d'),
  patientName: 'Priya',
  doctor: 'Dr. Kumar',
  department: 'Neurology',
  age: 48,
  treatmentCost: 27000,
  daysAdmitted: 5
}
{
  _id: ObjectId('6a115fab09c9f7fe9b192d6a'),
  patientName: 'Kiran',
  doctor: 'Dr. Kumar',
  department: 'Neurology',
  age: 40,
  treatmentCost: 25000,
  daysAdmitted: 6
}
```

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```
> db.patients.aggregate([
  {$limit: 5}
])
< {
  _id: ObjectId('6a115fab09c9f7fe9b192d66'),
  patientName: 'Arun',
  doctor: 'Dr. Ravi',
  department: 'Cardiology',
  age: 45,
  treatmentCost: 20000,
  daysAdmitted: 5
}
{
  _id: ObjectId('6a115fab09c9f7fe9b192d67'),
  patientName: 'Meena',
  doctor: 'Dr. Kumar',
  department: 'Neurology',
  age: 50,
  treatmentCost: 30000,
  daysAdmitted: 7
}
{
  _id: ObjectId('6a115fab09c9f7fe9b192d68'),
  patientName: 'Raj',
  doctor: 'Dr. Ravi',
  department: 'Cardiology',
  age: 60,
  treatmentCost: 15000,
  daysAdmitted: 3
}
```

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```
> db.patients.aggregate([
  {$skip: 3}
])
< {
  _id: ObjectId('6a115fab09c9f7fe9b192d69'),
  patientName: 'Divya',
  doctor: 'Dr. Mehta',
  department: 'Orthopedics',
  age: 35,
  treatmentCost: 10000,
  daysAdmitted: 2
}
{
  _id: ObjectId('6a115fab09c9f7fe9b192d6a'),
  patientName: 'Kiran',
  doctor: 'Dr. Kumar',
  department: 'Neurology',
  age: 40,
  treatmentCost: 25000,
  daysAdmitted: 6
}
{
  _id: ObjectId('6a115fab09c9f7fe9b192d6b'),
  patientName: 'Sneha',
  doctor: 'Dr. Mehta',
  department: 'Orthopedics',
  age: 30,
  treatmentCost: 8000,
  daysAdmitted: 2
}
```

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```
> db.patients.aggregate([
  {$match: { department: "Cardiology" }},
  { $project: {
    _id: 0,
    patientName: 1}}
])
```

```
< {
  patientName: 'Arun'
}
{
  patientName: 'Raj'
}
{
  patientName: 'Vijay'
}
{
  patientName: 'Riya'
}
```

```
> db.patients.aggregate([
  {$group: {
    _id: "$department",
    totalCost: { $sum: "$treatmentCost" }}}
])
```

```
< {
  _id: 'Orthopedics',
  totalCost: 30000
}
{
  _id: 'Cardiology',
```

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```
> db.patients.aggregate([
  {$group: {
    _id: "$doctor",
    averageCost: { $avg: "$treatmentCost" }
  }},
  {$sort: { averageCost: 1 }}
])
< {
  _id: 'Dr. Mehta',
  averageCost: 10000
}
{
  _id: 'Dr. Ravi',
  averageCost: 18750
}
{
  _id: 'Dr. Kumar',
  averageCost: 27333.333333333332
}
```

```
> db.patients.aggregate([
  {$group: {
    _id: "$department",
    totalCost: { $sum: "$treatmentCost" }}}},
  {$match: {
    totalCost: { $gt: 50000 }
  }}
])
< {
  _id: 'Neurology',
  totalCost: 82000
}
{
  _id: 'Cardiology',
  totalCost: 75000
}
```



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```
> db.patients.aggregate([
  {$group: {
    _id: "$department",
    patientCount: { $sum: 1 }}}},
  {$sort: { patientCount: -1 }}
])
```

```
< {
  _id: 'Cardiology',
  patientCount: 4
}
{
  _id: 'Neurology',
  patientCount: 3
}
{
  _id: 'Orthopedics',
  patientCount: 3
}
```

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```
> db.patients.aggregate([
  {$group: {
    _id: "$doctor",
    totalCost: { $sum: "$treatmentCost" }},
  {$sort: { totalCost: -1 }},
  {$limit: 3}
])
< {
  _id: 'Dr. Kumar',
  totalCost: 82000
}
{
  _id: 'Dr. Ravi',
  totalCost: 75000
}
{
  _id: 'Dr. Mehta',
  totalCost: 30000
}
```

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```
> db.patients.aggregate([
  {$match: {age: { $gt: 40 }}},
  {$group: {
    _id: "$department",
    patientCount: { $sum: 1 }}}
])
< {
  _id: 'Orthopedics',
  patientCount: 1
}
{
  _id: 'Cardiology',
  patientCount: 3
}
{
  _id: 'Neurology',
  patientCount: 2
}
```

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```
> db.patients.aggregate([
  {$group: {
    _id: "$department",
    averageCost: { $avg: "$treatmentCost" }}}},
  {$project: {
    _id: 0,
    department: "$_id",
    averageCost: 1}},
  {$sort: { averageCost: -1 }}
])
< {
  averageCost: 27333.333333333332,
  department: 'Neurology'
}
{
  averageCost: 18750,
  department: 'Cardiology'
}
{
  averageCost: 10000,
  department: 'Orthopedics'
}
```

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>

```
> db.patients.aggregate([
  {$group: {
    _id: "$department",
    patientCount: { $sum: 1 }}}},
  {$match: {patientCount: { $gt: 2 }}}
])
```

```
< {
  _id: 'Orthopedics',
  patientCount: 3
}
{
  _id: 'Cardiology',
  patientCount: 4
}
{
  _id: 'Neurology',
  patientCount: 3
}
```

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```
> db.patients.aggregate([
  {$group: {
    _id: "$department",
    totalCost: { $sum: "$treatmentCost" } }},
  {$sort: { totalCost: -1 }},
  {$skip: 1},
  {$limit: 3}
])
< {
  _id: 'Cardiology',
  totalCost: 75000
}
{
  _id: 'Orthopedics',
  totalCost: 30000
}
```

```
> db.patients.aggregate([
  {$match: {
    treatmentCost: { $gt: 10000 } }},
  {$group: {
    _id: "$doctor",
    totalCost: { $sum: "$treatmentCost" } }},
  {$sort: { totalCost: -1 }},
  {$limit: 2}
])
< {
  _id: 'Dr. Kumar',
  totalCost: 82000
}
{
  _id: 'Dr. Ravi',
  totalCost: 75000
}
```